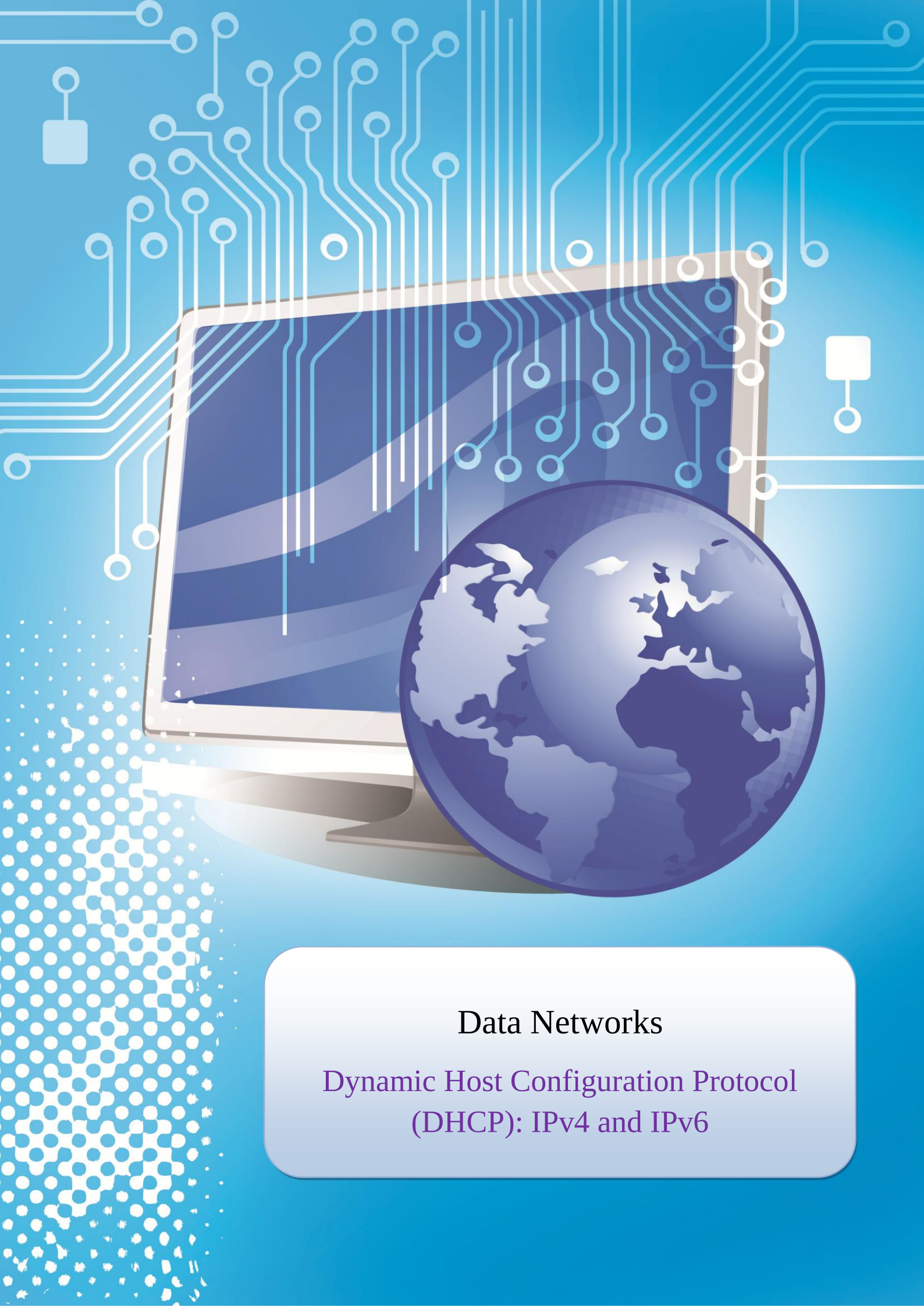




Data Networks

Dynamic Host Configuration Protocol
(DHCP): IPv4 and IPv6

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Keywords:

S.N	Keyword	Meaning
1	DHCP	Dynamic Host Configuration Protocol: protocol used to assign IP addresses to computers on a Microsoft NT local area network
2	IPV4	Internet Protocol version 4
3	IPV6	Internet Protocol version 6
4	Dual-stack	A term used to describe a network node running both IPv4 and IPv6 protocol stacks at the same time.
5	IANA	Internet Assigned Numbers Authority

Introduction:

DHCP is standard for host configuration protocol. And it is built upon version 4 of the IP, other name is IPv4. However, the second version of it is: IPv6. And we have two version of DHCP (DHCPv4 & DHCPv6). The First version of DHCP RFC 2131 designed for IPv4 and the second new version is DHCPv6 RFC 3315 for IPv6. We use DHCP to configure network easily, so we don't need to configure every single machine in the network manually. DHCP will push these configurations (IP, SUBNET, GETWAY) automatically to all machines in the LAN.

Internet Protocol version 4 (IPv4):

IPV4 was created in 1970, and based on 32 bit architecture to cover 4,294,967,296 (2³²) unique addresses. But, limitation of address allocation in IPv4 and there is no enough address and routers have unnecessary work to do and because of technology spreading in the all kind of networking and peripherals then the IPV4 is not able to cover all of these requests and we need to develop it to become more capable to cover more addresses. IPV6 was the solution.



IPv4 address consists of four parts, every part is byte or octet and we can write it like:

1. Using decimal: every part is separated by dot: 172.16.25.50
2. Using binary: 10101100.00010000.00011110.00111000
3. Using Hexadecimal: AC 10 1E 38

All of above is used to represent same address but in different ways, and we are using the decimal more than others.

IPv4 Special-use addresses:

In IPv4 there are some of reserved addresses blocks, this address can't be used because it is reserved by IANA for special use.

S.N	CIDR Address Block	Purpose
1	0.0.0.0/8	Used for broadcast messages to the current ("this") network
2	10.0.0.0/8	Used for local communications within a private network
3	127.0.0.0/8	Used for loopback addresses to the local host.
4	169.254.0.0/16	Used for autoconfiguration between two hosts on a single link when no IP address is otherwise specified
5	172.16.0.0/12	Used for local communications within a private network
6	192.0.0.0/24	Reserved (IANA)
7	192.0.2.0/24	Assigned as "TEST-NET" for use solely in documentation and example source code and should not be used publicly.
8	192.88.99.0/24	Used by 6to4 anycast relays
9	192.168.0.0/16	Used for local communications within a private network
10	198.18.0.0/15	Used for testing of inter-network communications between two separate subnets
11	198.51.100.0/24	Assigned as "TEST-NET-2" for use solely in documentation and example source code and should not be used publicly.
12	203.0.113.0/24	Assigned as "TEST-NET-3" for use solely in documentation and example source code and should not be used publicly.
13	224.0.0.0/4	Reserved for multicast assignments
14	240.0.0.0/4	Reserved (former Class E network)
15	255.255.255.255	255.255.255.255 is reserved for the "limited broadcast" destination address.

Internet Protocol version 6(IPV6):

IPv6 is a new version of IPv4, the growth of the internet needs more address than IPv4 capability. IPv6 has more numerical addresses than IPv4. IPV6 was created in 1998 based on 128 bit architecture to cover 2^{128} unique addresses. These expansions allow many devices and peripherals in allocating address. IPv6 addresses are written in hexadecimal, separated by colons instead of dots, grouped by double-bytes instead of bytes. But IPv6 still use CIDR notation for blocks of addresses.



IPv6 Special-use addresses:

S.N	CIDR Address Block	Purpose
1	::/128	Unspecified address
2	::1/128	Used for loopback address to the local host.

3	::ffff:0:0/96	IPv4 mapped addresses
4	2001::/32	Teredo tunneling
5	2001:10::/28	Overlay Routable Cryptographic Hash Identifiers (ORCHID)
6	2001:db8::/32	Addresses used in documentation
7	2002::/16	6to4
8	fc00::/7	Unique local address
9	ff00::/8	Multicast

How IP Addresses Assigned?

1. Dynamic Allocation.
2. Automatic Allocation.
3. Static Allocation.

Network classes:

1. Class A: Less networks, More hosts
2. Class B: Networks and hosts are medium.
3. Class C: More networks, less hosts

Who is affected by the Transition to IPv6?

In the future, no more IPv4 addresses issued. All addresses have to be IPv6. This problem is affect everyone with existing IPv4 addresses today and everyone require an IP address in the future. And because of growth in mobile devices, sensors and computers, IPv6 will be more important. This conversion to IPv6 will not happen overnight. The transition will be gradual over the next 10-15 years. Commercial and users worldwide will need to interoperate between IPv4 and IPv6 until all IPv4 devices are converted or retired

DHCP History:

1. RFC 1531: DHCP is defined as standard protocol in 1993.
2. RFC 2131: DHCP is upgraded to be appropriate with IPV4.
3. RFC 3315: DHCP is upgraded to be appropriate with IPV6.

Benefits of using DHCP Server:

1. Assign IP Address automatically to every single PC which log in to the network.
2. Configure TCP/IP protocol to all clients who are using DHCP service.
3. If we don't have DHCP server, we should do the configuration to every PC manually.

Differences between DHCPv6 and DHCPv4:

1. In DHCPv6 IPv6 address allocation is handled using a message option.
2. The message types like DHCPDISCOVER and DHCPOFFER supported by DHCPv4 are removed in DHCPv6. Instead, DHCPv6 servers are located by a client SOLICIT message followed by a server ADVERTISE message
3. DHCPv6 clients can request multiple addresses, which is not included in DHCPv4 clients.

Differences between DHCP for IPv4 and IPv6:

- Hosts always have a link local address that can be used in requests (in IPv4 0.0.0.0 is used as source address).
- Uses special multicast addresses for relay agents and servers.
- No compatibility with BOOTP, since no BOOTP support on IPv6.
- Simplified two-message exchange for simple configuration cases.
- A client can request multiple IPv6 addresses.
- Client can send multiple unrelated requests to the same or different servers
- There is a reconfigure message where servers can tell clients to reconfigure. This feature is optional.

Difference between DHCPv6 and DHCPv4 in IPV4:

- **No baggage :**
 - DHCP4 is based on BOOTP protocol. This packet layout is wasteful in many cases. Due to backward compatibility, there are a number of restrictions that further complicate DHCP for IPv4
 - A lot of the options turn out to be not useful, or not as useful as they can be, but it is hard to change a protocol with such a large installed base.
 - There are a lot of "tweaks" that implementations need in order to be compatible with the buggy clients.
- **IPv6 is better. Four features of IPv6 greatly improve DHCPv6:**
 - IPv6 hosts have "link-local addresses". Every network interface has a unique address that can be used to send and receive on the link only. IPv6 hosts can use this to send requests for "real" addresses. IPv4 hosts have to use system-specific hacks to work before they have an address.
 - All IPv6 systems support multicasting. All DHCPv6 servers register that they want to receive DHCPv6 multicast packets. This means the network knows where to send them. In IPv4, clients broadcast their requests, and networks do not know how far to send them.
 - DHCPv6 supports robust relaying, with up to 32 relays in each network. This greatly improves scalability and network manageability.

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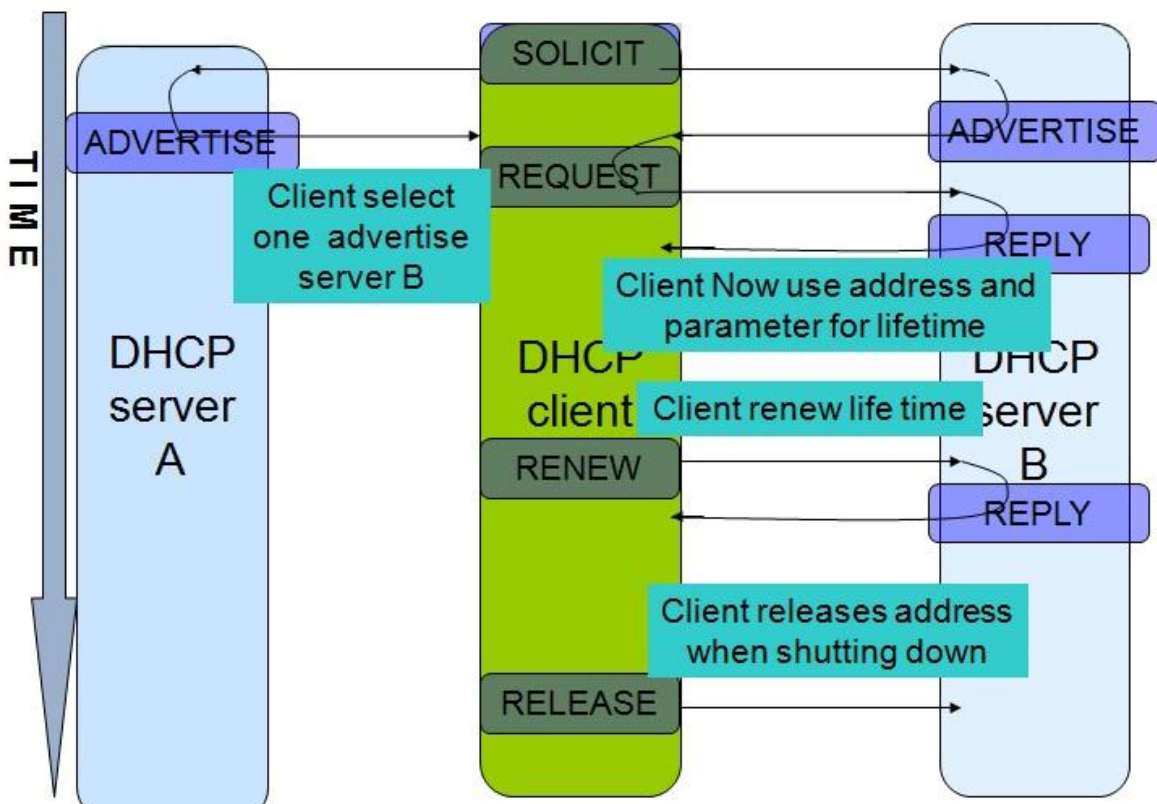
- Thanks to number of features like deprecated addresses and information refresh time option, network administrators can easily perform configuration changes and even network renumbering with users barely noticing.

The Needing for DHCPV6:

1. Providing IPv6 address configuration/assignment, DNS servers and NIS information
2. DHCP allows centralized control and auditing of IP address assignments:
 - Central point of control for resources
 - Dynamic DNS updates.
 - More security.

How DHCPv6 works:

1. Client sends messages to link-local multicast address
2. Server unicasts response to client
3. Information-Request / Reply - provide client configuration Information, but no addresses
4. Confirm / Reply - assist in determining whether client moved
5. Reconfigure - allow servers to initiate a client reconfiguration
6. Basic client/server authentication capabilities in base standard.
7. DHCP Unique Identifier (DUID) used to identify clients & servers
8. Identity Association ID (IAID) used to identify a collection of addresses
9. Relay Agents used when server not on-link
10. Relay Agents may be chained



DHCPv6 Protocols and Addressing:

1. DHCPv6 operation is similar to DHCPv4. The difference is the protocol itself has been completely rewritten.
2. It is not based on the older DHCPv4 or BOOTP, except in conceptual terms.
3. It still uses UDP but uses:
 - New port number
 - A new message format
 - Restructured options.

IPv4 VS IPv6:

In 1998, IPv4 has been verified, after that some changes happened on the IPv4 system. IPv4 consist of four parts as: 00000000.00000000.00000000.00000000 and every part consist of eight binary digits, every digit is either 0 or 1. therefore, every part has capacity between zero to 256. And we have 32 digits that mean there are 4,294,967,296 numbers to use. There are 18 million number are reserved for private networks, 270 million for multicasting to solve some technical problems. The world has been divided into five parts and every part got a range of numbers for every region, it's called RIR-Regional Internet Registry, the last distribution for the Asia Pacific that consumed by the end of the third day of February 2011.



After that, companies used IPv6 which is designed to replace the old system when the numbers reach only the top, IETF did the job since 1998. The system consists of the new 128-Digit, i.e., the upper limit of the numbers are 2128 (340 undecillion or 3.4×10^{38} , and it's very large number to covers human beings ,especially as the punctuation in letters and numbers on a system hexadecimal from 0 to 15, but after the number 9 becomes ABCDEF.

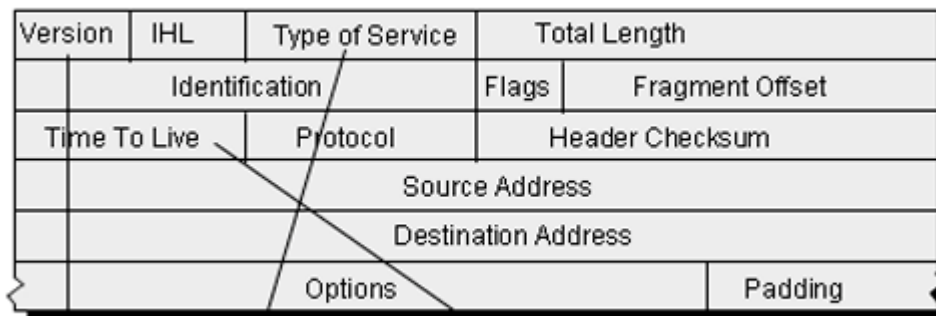
During transition, there are some difficulties to deal with new system. In this situation, there are millions of numbers still exist, these number are divided for two kind, first one is constant for websites and servers, and other for non-permanent users. Most of them doesn't stay on the network forever, therefore the new user can get the number which is belong to other user in the past.

IPv6 has ability to deal with billion of terminals, decrease the size of routing table, simplify the protocol just to allow router to be faster when processing the packets, better security than IPv4, allow protocol to be developed in future.

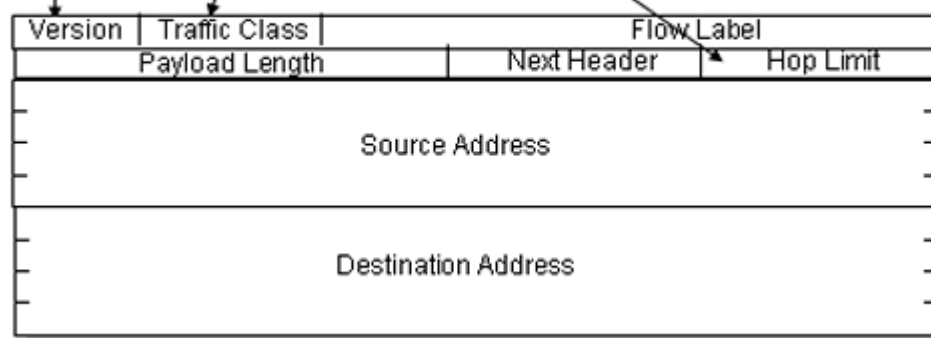
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S.N	IPv4	IPv6
1	IP uses 32 bit or 4 byte	IP uses 128 bit or 16 byte
2	Produce 4,294,967,296 IPs	Produce 340,282,366,920,938,463,374607,431,768,211,456 IPs
3	Header has Checksum	Checksum has been removed from the header, and replaced by checksum in link layer
4	Header has option	Option has been removed from the header, and replaced by extension header which is gave a space to add some options.
5	Not supported	New technology (Anycast) to speed the data transition
6	Uses Arp Protocol to get mac address for specific IP	Arp Protocol has stopped and replaced by Multicast Neighbor Solicitation
7	Uses IGMP protocol for Multicast Group	Uses MLD protocol for Multicast Group
8	No space for Flow Label and Qos is processing through router	Flow Label has been added to the Header especially for Qos
9		

IPv4 Header



IPv6 Header



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